Nanru (Anna) Dai

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EDUCATION

University of Pennsylvania, Philadelphia, PA **Aug 2026 – May 2028**
M.S.E. in Computer Graphics and Game Technology GPA: 4.0/4.0
Relevant Coursework: GPU Programming, Computer Animation, Procedural Computer Graphics

Johns Hopkins University, Baltimore, MD **Aug 2022 – May 2026**
B.S. in Computer Science & Applied Mathematics and Statistics, Minor in Robotics GPA: 3.77/4.0
Relevant Coursework: Computer Graphics, 3D Game Programming, Deep Learning, Operating Systems

SKILLS

Languages: C++, C, Python, Java, TypeScript, GLSL, HLSL, WGSL, VEX, Assembly, Lua, Bash
Graphics & GPU: CUDA, Vulkan, WebGPU, WebGL2, Metal, Raylib
DCC & Engines: Houdini (HDAs, Vellum, Karma/Solaris), Unity (URP, Shader Graph), Blender
Tools: Git, Nsight Systems, Nsight Compute

PROJECTS

CUDA Path Tracer | *CIS 5650 GPU Programming* Sep 2026 – Oct 2026
- Built a physically based path tracer parallelized over path segments, with stream compaction, material sorting, and a from-scratch glTF 2.0 loader feeding a per-mesh BVH (up to 21× faster); profiled with Nsight Systems.
- Implemented Fresnel glass with total internal reflection, image and procedural textures with bump mapping, Phong glossy reflection, direct light sampling (10× less noise at depth 2), and thin-lens depth of field.

WebGPU Forward+ & Clustered Deferred Shading | *CIS 5650 GPU Programming* Oct 2026
- Rendered Sponza with 5,000 moving lights using compute-shader light clustering in WebGPU/WGSL, cutting frame time from 352 ms (naive) to 29.5 ms with Forward+ and 9 ms with clustered deferred shading.
- Added an alpha-tested depth prepass that cut Forward+ from 28.1 to 12.0 ms at 5,000 lights, and caught invalid benchmark results caused by GPU downclocking and per-cluster light-cap overflow.

Vulkan Grass Rendering | *CIS 5650 GPU Programming* Oct 2026 – Nov 2026
- Simulated up to 1M Bézier grass blades in Vulkan compute shaders with gravity, recovery, and wind, appending survivors to a GPU-side indirect draw; orientation, frustum, and distance culling raised 1M blades from 24 to 95 FPS.
- Cut tessellated triangles per blade from 162 to 18 by removing width subdivision that never changed the blade's shape, nearly doubling frame rate at 1M blades (85 to 177 FPS), with distance-based tessellation LOD on top.

Jiangnan Procedural Buildings: Houdini HDA | *CIS 5660 Procedural Computer Graphics* Oct 2026 – Nov 2026
- Built a Houdini building HDA where one tool produces both a stilted two-floor teahouse and a three-floor restaurant from per-floor parameters, with a VEX layout routine that re-spaces windows, doors, and pillars from asset sizes on all four walls.
- Modeled Jiangnan details as nested HDAs (shared lattice panels, 2/4/6-leaf lattice doors, leaning-seat railings, overhanging-gable roof, lanterns), an Auto support mode choosing brackets or stilts by overhang depth, and a canal scene.

Procedural Jellyfish | *CIS 5660 Procedural Computer Graphics* Oct 2026
- Built an animated jellyfish in Houdini around a single keyframed bell controller: veins and organs follow through Point Deform against a frozen rest pose, and Vellum arms and tentacles are pinned to the moving bell.
- Grew branching veins as shortest paths over the remeshed bell using a noisy VEX cost field, and shaded the creature in Karma with MaterialX transmission, subsurface scattering, and emission.

Balloon Bunny: Stylized Toon Scene | *CIS 5660 Procedural Computer Graphics* Sep 2026 – Oct 2026
- Recreated a 2D illustration as a stylized Unity URP scene with a Shader Graph/HLSL toon shader: three-tone ramp across multiple lights, hand-drawn hatching sampled in UV space, and toggleable rim, specular, and edge-glint highlights.
- Built full-screen passes for Roberts Cross depth/normal outlines with stepped-noise "line boil", a paper texture, and a depth-masked sunset sky, plus a one-key style switch that swaps materials and effects scene-wide.

EXPERIENCE

Johns Hopkins University | *Course Assistant, Computer Graphics & Math Foundations* Aug 2024 – Dec 2025

Google, New York | *STEP Intern, Mendel Lifecycle Team* May 2024 – Aug 2024
- Developed Angular and TypeScript components for an internal experimentation platform, streamlining experiment setup and monitoring.
- Built a template-based configuration tool automating eligibility checks and pre-populating setup forms, reducing manual setup time and onboarding effort by over 50%.

Microsoft, Beijing, China | *Software Engineering Intern, AI Vertical Group* Jun 2023 – Jul 2023
- Built an NLP pipeline with SpaCy NER and GPT APIs to extract character relationships, visualized with NetworkX.
- Enhanced a document chatbot by optimizing LlamaIndex retrieval and building document management features, improving response accuracy and latency.

RESEARCH

World-in-World: Building a Closed-Loop World Interface to Evaluate World Models Mar 2025 – Oct 2025
ICLR 2026 Oral | *Advisor: Prof. Alan Yuille, Johns Hopkins University*
- Designed and conducted experiments for a closed-loop evaluation framework measuring reasoning and goal completion of generative world models beyond visual fidelity.
- Built demo pipelines integrating generative models across four simulated environments.